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Development of Autonomous Rover with Robotic Arm for Sorting Based on Color

Project Focus: Development of a semi-autonomous rover with an integrated robotic arm for efficient color-based sorting.

Object Detection & Classification: Uses RGB sensors and a robotic arm for accurate and automated sorting.

Application Areas: Suitable for industrial and agricultural environments.

Control Mechanism: Bluetooth-enabled control allows both manual and automated operation, enhancing flexibility.

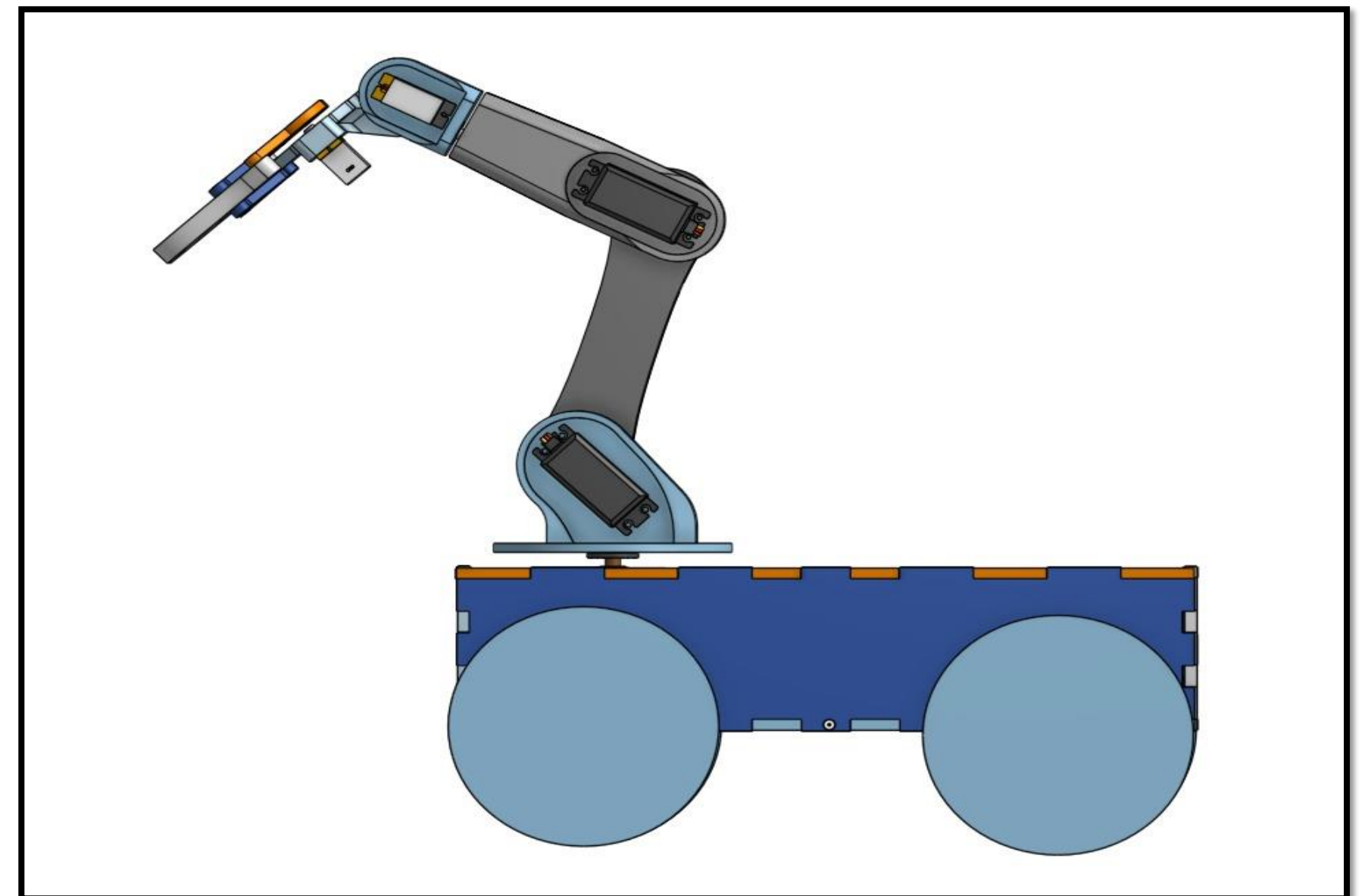
Benefits:

- Reduces manual labor.
- Minimizes sorting errors.
- Enhances productivity.

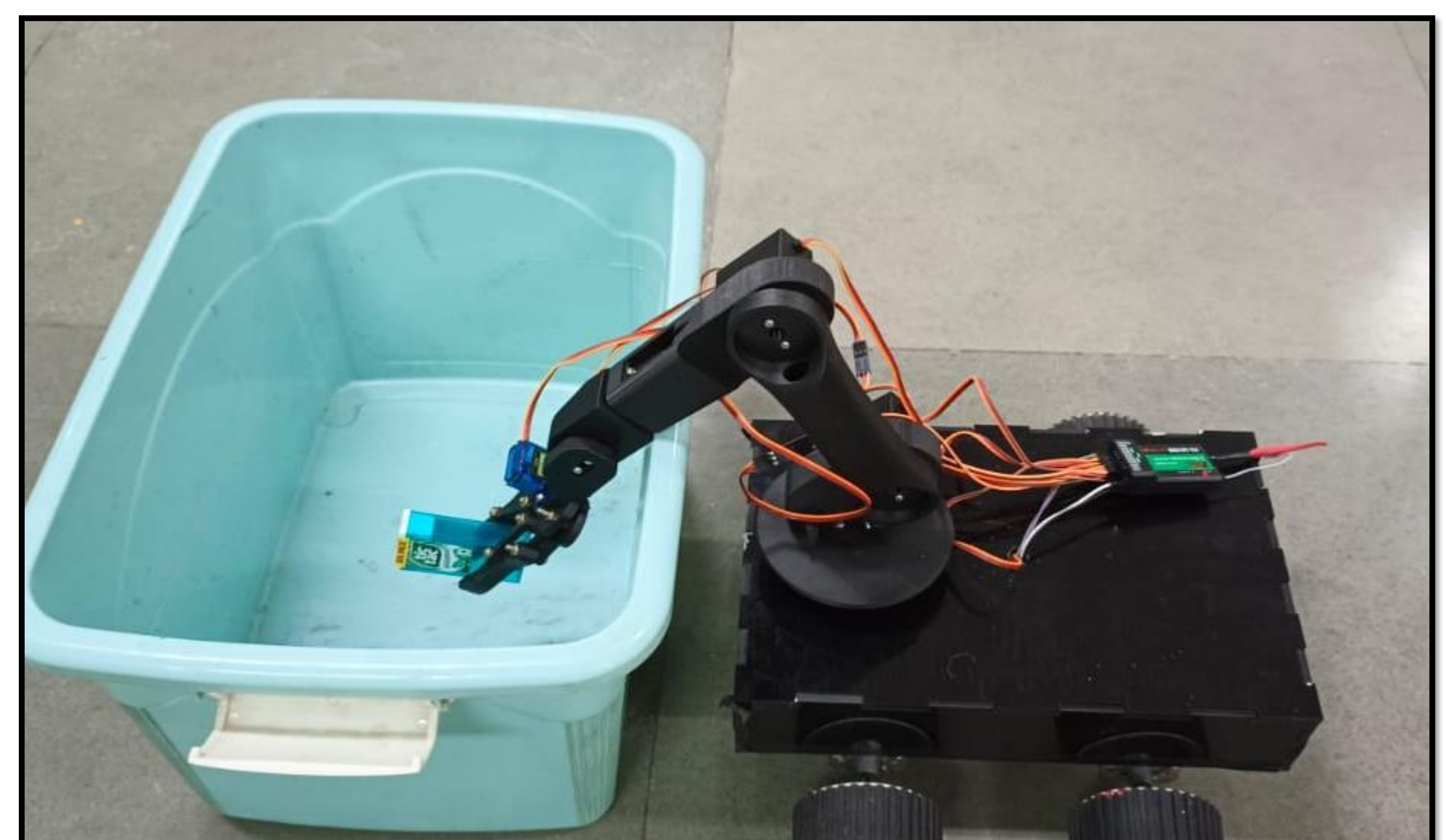
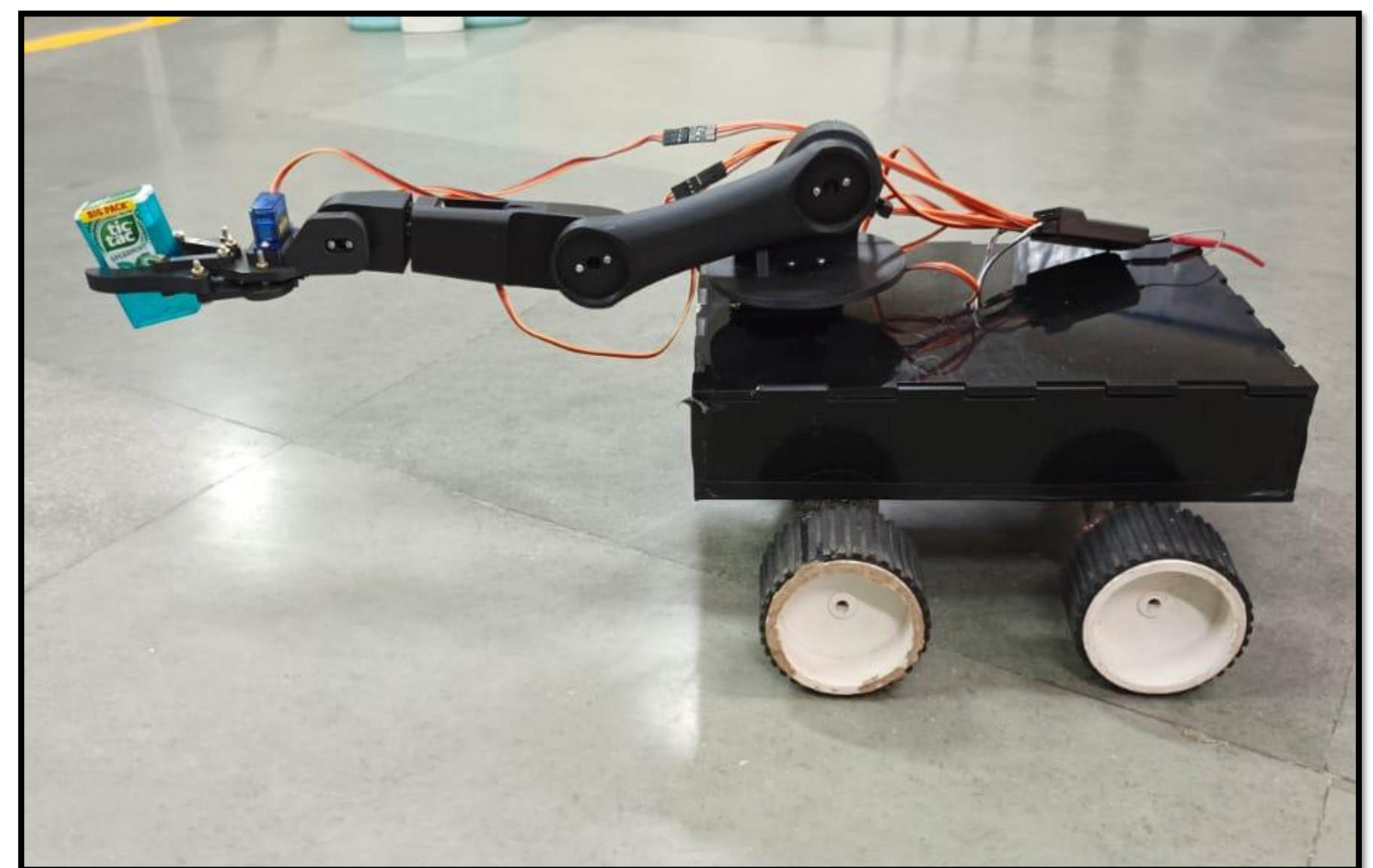
Future Advancements:

- Integration of machine learning for improved object detection.
- Enhanced adaptability in dynamic environments.

Impact: Contributes to automation technology, optimizing efficiency and resource management across multiple industries.



CAD Model



Pick and Place Operations

Academic Guide

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Team Members

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